

III-B.Tech I-Semester Regular Examinations, February-2022.

COMPUTER NETWORKS
(Computer Science and Engineering)

Max. Marks: 70

Time: 03 Hours

(Answer any 05 full questions. Each question carries 14 marks)

Q.No.	Question	Marks	Bloom's Level
Q.1(a)	Illustrate with figure OSI model and in detail explain the functions of OSI layers.	[08]	[L2]
(b)	Identify the different network topologies and explain them.	[06]	[L3]
Q.2(a)	Explain different propagation modes in optical fiber and discuss its advantages and disadvantages.	[08]	[L2]
(b)	Differentiate between micro waves and radio waves.	[06]	[L4]
Q.3(a)	What are the different framing methods used in data link layer? Explain them in detail.	[08]	[L1]
(b)	An 8-bit binary value 10101111 is to be encoded using an even parity Hamming code. What is the binary value after encoding?	[06]	[L3]
Q.4(a)	Describe CSMA-CD protocol.	[08]	[L1]
(b)	How good are limited contention protocols in handling delay and channel efficiency? Discuss.	[06]	[L4]
Q.5(a)	Draw IPV4 packet size header format and explain relevance of each field.	[08]	[L2]
(b)	Demonstrate the Link state routing with an example.	[06]	[L3]
Q.6(a)	Explain operation of ARP protocol.	[08]	[L2]
(b)	Describe the role of home agent and foreign agent in mobile IP.	[06]	[L1]
Q.7(a)	Explain TCP Connection establishment with a neat diagram.	[08]	[L2]
(b)	Draw UDP header format and explain with an example.	[06]	[L2]
Q.8(a)	Illustrate with the help of necessary diagrams, the general architecture of Email system.	[08]	[L2]
(b)	Categorize web documents and discuss the languages used to create those web documents.	[06]	[L4]

END OF THE QUESTION PAPER